

DEPTH EVIDENCE: UNIFYING METRIC DEPTH PREDICTION AND GEOMETRIC REASONING IN MULTIMODAL LANGUAGE MODELS

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ABSTRACT

Spatial reasoning with metric constraints requires linking objects to geometric measurements and preserving their numerical content during language reasoning. We present DepthEvidence, a 4B model that uses its own dense metric predictions as object-grounded evidence for language generation. A camera-conditioned decoder predicts full-resolution metric depth using multi-scale visual features and high-resolution RGB refinement. A dense-to-language interface converts predicted depths and decoder features into object-aligned continuous geometry tokens anchored to object identifiers. Geometric supervision encourages metric information to remain recoverable before and after language-context interaction, while instruction tuning supports object measurement and compositional reasoning. We introduce a Depth-VQA benchmark evaluating object-depth queries, relative comparisons, and decisions combining spatial and numerical constraints. Across nine datasets, DepthEvidence achieves the highest average dense δ_1 among evaluated methods, competitive with specialized estimators. It also leads the evaluated methods in instance-level metric depth estimation and overall accuracy on both relative and metric reasoning tracks, while broadly preserving general VQA performance and improving spatial understanding relative to the base model.

1 INTRODUCTION

Answering spatial questions with metric constraints requires connecting object semantics to geometric measurements. Knowing which object is closer establishes a relative relation, but selecting objects within a depth interval requires absolute scale and the correct association between each object and its measured depth. Consider the question in Figure 1(a): *which referenced objects to the right of a wooden cabinet have an average camera depth between 1.30 m and 1.33 m?* The answer must satisfy both the spatial filter and the numerical interval. Dense metric estimators provide pixel-level measurements (Yin et al., 2023; Piccinelli et al., 2024), while spatially aware multimodal large language models (MLLMs) connect objects with language (Chen et al., 2024a; Cheng et al., 2024). This question requires bringing those capabilities together at the level of individual objects.

Recent depth-aware VLMs have substantially advanced geometric prediction. DepthLM (Cai et al., 2026) demonstrates accurate pixel-level metric estimation through language modeling, using visual prompting and intrinsic-conditioned augmentation without a dedicated depth head. DepthVLM (Yu et al., 2026) extends this capability to full-resolution dense prediction through a lightweight head and joint depth-text training, while retaining general multimodal capability and improving spatial reasoning. G²VLM also couples geometric reconstruction with spatial reasoning (Hu et al., 2026). Building on these advances, we ask: *how can a VLM use its own dense metric predictions as object-grounded evidence for language reasoning?* We focus on explicitly linking the predicted metric field to the referenced objects and preserving its numerical content as the language model processes the question.

Using predicted geometry as language evidence requires both object alignment and metric information preservation. Measurements must come from the appropriate image regions and remain associated

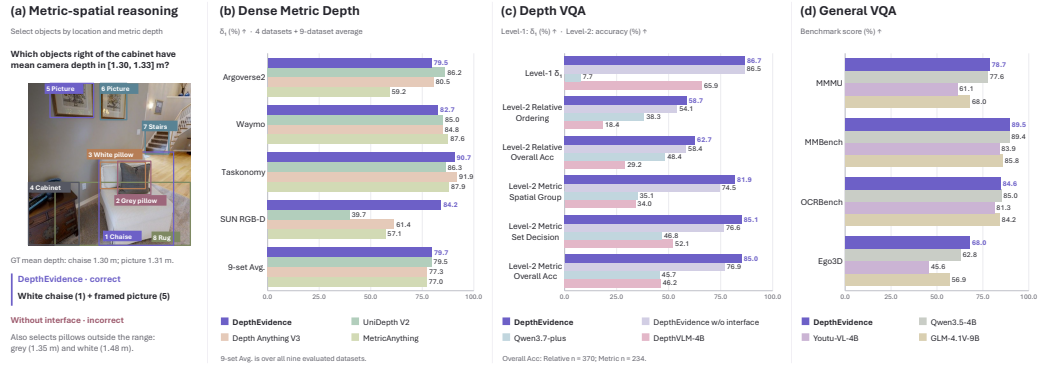


Figure 1: Overview of DepthEvidence’s capabilities. (a) A metric-spatial question requires selecting objects by image location and mean camera depth; the dense-to-language interface rejects out-of-range distractors. (b) Dense metric depth results (δ_1) on four representative datasets and the nine-dataset average, compared with specialized estimators. (c) Level 1 metric estimation and Level 2 relative and metric reasoning, compared with general-purpose and depth-aware VLMs as well as the no-interface variant. (d) General VQA retention on MMMU, MMBench, and OCRBench, and spatial understanding on Ego3D. All scores are percentages and higher is better.

with the correct references as the model processes multiple objects. Their numerical content must also survive projection into language tokens and interaction with the question. Answer supervision targets the final response without directly constraining the metric content of intermediate geometry tokens. This motivates explicit geometric supervision to keep object-level depth information recoverable before and after interaction with the language context.

We present DepthEvidence, a 4B model that turns its own dense predictions into object-aligned evidence for language generation. Given an image and camera intrinsics, a camera-conditioned decoder combines multi-scale visual features with high-resolution RGB refinement to produce metric depth and preserve local structure. The dense-to-language interface pools predicted depths and decoder features within each referenced region, then encodes them as object-aligned continuous geometry tokens anchored to object identifiers. Auxiliary depth objectives supervise metric information in these tokens before and after language-context interaction, while a ranking objective supervises object ordering. The tokens enter answer generation through cached continuation, avoiding repeated encoding of the image and question. In the cabinet example, this interface supplies the candidate objects’ geometry for spatial filtering and metric-range selection.

To evaluate how geometric prediction translates into language answers, we organize depth understanding into dense metric depth prediction (Level 0), instance-level metric depth estimation (Level 1), and compositional depth reasoning (Level 2). Instance-level instruction tuning and the interface support both measurement and reasoning. Our Depth-VQA benchmark evaluates whether models can turn object-level depth measurements into compositional answers, covering relative comparisons and decisions that jointly depend on spatial relations and metric gaps, ranges, or thresholds. DepthEvidence achieves the highest average dense δ_1 across nine datasets among the evaluated methods and the highest overall accuracy on both reasoning tracks, while general VQA performance remains broadly stable relative to the base model (Figure 1).

Our contributions are:

- An object-aligned dense-to-language interface with geometric supervision before and after contextualization, enabling language generation to condition on the model’s own metric predictions.
- DepthEvidence, a unified 4B model combining camera-conditioned dense prediction with instance-level depth instruction tuning, together with a Depth-VQA benchmark that distinguishes object measurement, relative reasoning, and metric-constrained reasoning.
- DepthEvidence leads the evaluated methods in nine-dataset average dense δ_1 , instance-level metric estimation, and overall compositional reasoning, while retaining general VQA performance and improving spatial understanding over the base model.

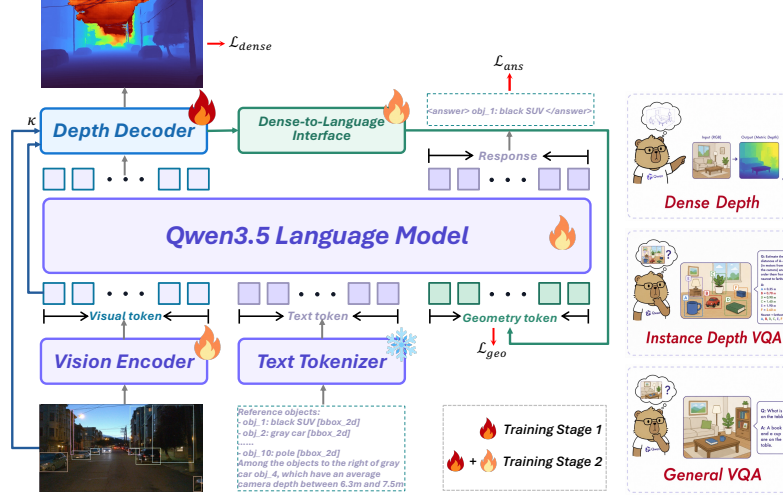


Figure 2: Overview of DepthEvidence. The dense decoder supplies metric predictions and features to the dense-to-language interface, which converts them into object-aligned geometry tokens for answer generation. This path connects Level-0 dense prediction to Level-1 instance measurement and Level-2 compositional reasoning within a shared VLM.

2 RELATED WORK

Monocular Metric Depth Estimation. Monocular depth estimation has progressed from multi-scale convolutional models with scale-invariant supervision (Eigen et al., 2014) to transformer decoders using multi-level visual tokens (Ranftl et al., 2021). Depth Anything and Depth Anything V2 scale transferable representations with unlabeled or pseudo-labeled data (Yang et al., 2024a;b), whereas Metric3D and UniDepth address camera variation through canonicalization or camera-conditioned features (Yin et al., 2023; Piccinelli et al., 2024). These methods primarily treat geometry as a dense output; DepthEvidence additionally uses metric predictions as object-grounded evidence for measurement and reasoning.

Geometry-Aware Vision–Language Models. Prior work uses geometry either to support spatial reasoning or as an explicit model output. SpatialVLM constructs metric spatial question–answer data through automatic 3D annotation (Chen et al., 2024a), and SpatialRGPT combines 3D scene-graph regions with a depth-aware visual plugin (Cheng et al., 2024); G²VLM, Vid-LLM, and SpatialStack more directly couple reconstruction, 3D tokens, or geometry-encoder features with language reasoning (Hu et al., 2026; Chen et al., 2026; Zhang et al., 2026). AURORA and Perceptio emit discrete depth-related tokens before text (Bigverdi et al., 2025; Li et al., 2026), DepthLM predicts metric depth at prompted pixels (Cai et al., 2026), and DepthVLM jointly outputs full-resolution depth and text (Yu et al., 2026). However, explicit or parallel depth prediction does not necessarily condition answers on the predicted metric field. DepthEvidence bridges prediction and reasoning by converting its single-image dense outputs into object-aligned continuous geometry tokens, supervised before and after language-context interaction, for object-level measurement and relative or metric compositional reasoning.

3 METHOD

3.1 FRAMEWORK OVERVIEW

As illustrated in Figure 2, DepthEvidence conditions object-level answers on geometric evidence predicted from the same image. A camera-conditioned dense decoder supplies a metric depth map and spatial features. The dense-to-language interface extracts evidence for objects specified by boxes or referring prompts and binds their geometry tokens to explicit object identifiers. Auxiliary objectives supervise metric-depth prediction from these tokens before and after language-context interaction. Instance-level instruction tuning trains the VLM to use this context for measurement and compositional reasoning.

3.2 LEVEL-0 DENSE METRIC DEPTH PREDICTION

Camera-conditioned decoder. The Level-0 branch follows a coarse-to-fine DPT design (Ranftl et al., 2021), fusing intermediate vision-encoder features with the spatially reshaped final visual hidden states. To recover metric scale, it converts the supplied camera intrinsics κ into a compact descriptor containing normalized calibration, field-of-view, and aspect-ratio information. An MLP maps this descriptor to FiLM parameters (Perez et al., 2018) that condition every decoder scale. A lightweight high-resolution RGB pathway supplies finer appearance cues during upsampling, and the resulting full-resolution representation predicts both the metric depth map and the dense features used by the dense-to-language interface. This numerical camera encoding is specific to the dense branch; Depth-VQA instead receives the raw intrinsic values in its instruction. Appendix A.1.1 provides the feature dimensions, camera parameterization, and decoder details.

High-resolution RGB refinement. Spatially compressed visual tokens can lose thin structures and sharp depth discontinuities. The RGB pathway therefore injects high-resolution features through learned residual gates, preserving the transformer representation as the main path while restoring local boundary cues. The refined spatial features also provide local appearance and boundary cues for object-level pooling in the interface. Architectural details are given in Appendix A.1.2.

Training objective. We train the dense branch with

$$\mathcal{L}_{\text{dense}} = 0.8\mathcal{L}_{\text{abs}} + 0.2\mathcal{L}_{\text{silog}} + 0.1\mathcal{L}_{\text{der}} + 0.03\mathcal{L}_{\text{lap}} + \mathcal{L}_{\text{ssi}}. \quad (1)$$

The first two terms supervise metric depth, the derivative and Laplacian terms preserve multi-scale structure, and the edge-guided stable local scale-and-shift invariant term $\mathcal{L}_{\text{ssi}}$ emphasizes object boundaries. Appendix A.1.3 provides the detailed definitions and implementation.

3.3 INSTANCE-LEVEL DEPTH INSTRUCTION TUNING

Instance-level measurement and reasoning tasks. We extend dense prediction with two object-grounded instruction levels. Level 1 asks for a specified object’s center, mean, median, nearest, or farthest metric depth. Level 2 combines measurements across objects: relative tasks cover near–far and front–behind comparisons, closest/farthest selection, and subset or full depth ordering; metric tasks cover numerical mean-depth gaps, thresholds, ranges, and margin-based decisions. Compositional questions first resolve image-space references or filter candidates, then apply relative comparisons or metric constraints. These instructions teach object-level measurement and its use in spatially and numerically constrained answers.

Unified object grounding. We support two ways to specify an object. A box prompt provides its category and bounding box, with coordinates normalized to a 0–1000 image coordinate system. A referring prompt instead identifies it through a natural-language description enclosed by object-reference tokens. For multi-object questions, each instance receives a stable identifier such as `obj_1`, and the prompt lists its identifier, category, and grounding cue before the question. This common serialization supports both grounding modes and disambiguates repeated object categories without assuming any interface-specific processing.

Structured supervision. Each instruction supplies the raw camera intrinsics as `<focal> fx fy cx cy </focal>`, independently of the dense decoder’s numerical camera encoding. The VLM reads this block together with the image and question. We train both instruction levels with the next-token cross-entropy loss $\mathcal{L}_{\text{ans}}$ on the final answer. The answer is a metric depth value for Level 1, or a numerical depth gap, an object or set selection, an ordering, or a discrete comparison or constraint decision for Level 2. Margin-based decisions include an *ambiguous* answer when the closest object does not satisfy the required depth margin. Appendix A.2 provides the detailed targets, prompt serialization, and output formats.

3.4 DENSE-TO-LANGUAGE INTERFACE

Object-aligned dense geometry. For both Level 1 and Level 2 instructions, the interface receives a metric depth map $\hat{\mathbf{D}}$ and a full-resolution feature map $\mathbf{F}^d$ from the dense decoder. For a box prompt, we use the supplied region directly; for a referring prompt, a lightweight grounding head predicts its

box. For object i , pooling $\mathbf{F}^d$ yields a regional feature $\mathbf{r}_i$, while summarizing the predicted depths yields metric statistics $\mathbf{s}_i$. Together with the normalized box $\mathbf{b}_i$, these form an object descriptor that associates local geometry with the referenced instance. Appendix A.3.1 gives the pooling statistics and grounding-head implementation.

Continuous geometry tokens. A two-layer projector P_g maps each object descriptor to $K = 2$ continuous tokens,

$$\mathbf{g}_{i,k} = P_g([\mathbf{r}_i; \mathbf{b}_i; \mathbf{s}_i])_k + \mathbf{p}_i + \mathbf{t}_k, \quad k \in \{1, 2\}, \quad (2)$$

where $\mathbf{p}_i$ is a learned object-index embedding and $\mathbf{t}_k$ distinguishes the two token slots. We organize the resulting tokens as an input-only geometry bank in which each pair follows its textual anchor `obj_i`. Geometry-bank positions are excluded as language-modeling targets, but answer tokens attend to them and the answer loss backpropagates through them to the projector. Appendix A.3.1 details the bank construction, and Appendix A.3.2 discusses the choice of $K = 2$.

Geometric supervision. To encourage metric-depth recoverability, we supervise the geometry representations at both instance levels with

$$\mathcal{L}_{\text{geo}} = 0.2\mathcal{L}_{\text{raw}} + 0.2\mathcal{L}_{\text{ctx}} + 0.1\mathcal{L}_{\text{rank}}, \quad (3)$$

The raw-depth and contextual-depth losses train auxiliary heads to predict five ground-truth object-depth statistics (center, mean, median, nearest, and farthest) in log space, from the projected tokens and their contextualized states, respectively. These objectives encourage metric-depth recoverability before and after interaction with the image and question. For multi-object examples, $\mathcal{L}_{\text{rank}}$ supervises relative ordering using the contextual mean-depth predictions. Appendix A.3.4 specifies the heads, targets, and gradient routing.

Single-prefill cached continuation. The geometry bank becomes available after the image and question have been encoded. We retain the prefix cache, append the bank as a causal continuation, and generate the answer from the extended context. Geometry tokens can attend to the cached prefix, and answer tokens can attend to both the prefix and geometry. This execution scheme incorporates predicted evidence without re-encoding the multimodal prefix. Appendix A.3.3 gives implementation details, and Appendix E compares no-interface, two-pass, and single-prefill inference.

4 INSTANCE-LEVEL DEPTH-VQA BENCHMARK

Depth-VQA evaluates object-level metric estimation (Level 1) and multi-object compositional reasoning (Level 2), including relative comparisons, ordering, numerical gaps, and spatially conditioned metric decisions. Each question provides an image, camera intrinsics, and object references specified by normalized category-box prompts or natural-language referring prompts.

Construction and quality control. We first complete sparse outdoor depth so every source image has a dense depth map. Qwen3.7-plus (Qwen Team, 2026b) and Seed-2.1-Turbo (ByteDance Seed Team, 2026) independently identify major objects and propose their categories and grounding boxes, reducing dependence on the output distribution of a single generator. We pool their candidates, remove boxes covering less than 1% of the image or lying near image boundaries, prompt SAM 3 (Carion et al., 2026) with the remaining categories and boxes, and apply a second quality filter to the resulting masks. Each retained mask and dense depth map yield the Level 1 center, mean, median, nearest, and farthest targets, with the latter two defined by the 5th and 95th depth percentiles. The two generators then produce Level 2 questions and answers from these instance records, with each candidate verified by the model that did not generate it. We manually audit 1% of training samples across all three levels to derive targeted filters, manually check every benchmark example, and limit reuse of the same instance across task types within a video scene. Appendix C.1 provides full details.

Level-1: Instance-Level Metric Depth Estimation. Level 1 contains 757 grounded instances from six indoor and outdoor datasets: Waymo (Sun et al., 2020), Argoverse 2 (Wilson et al., 2021), Taskonomy (Zamir et al., 2018), LingBot-Depth (Tan et al., 2026), DDAD (Guizilini et al., 2020), and nuScenes (Caesar et al., 2020). Each query requests the object-center, nearest, farthest, mean, or median valid depth within the referenced mask and expects a scalar answer in meters. We report δ_1 (higher is better), RMSE, and AbsRel (lower is better).

Table 1: Level-2 tracks and subbench goals. Appendix Table 7 provides all question types and reasoning requirements.

Track	Subbench	Question summary
Relative Depth	Spatial Pair	Compare two objects, optionally after resolving a spatial reference.
	Spatial Group	Spatially filter candidates, then select the nearest or farthest.
	Depth Ordering	Select an extreme or order a subset or all referenced objects.
Metric Depth	Metric Pair	Predict a mean-depth gap or compare it with a threshold.
	Spatial Metric Pair	Resolve a spatial reference, then predict an absolute mean-depth gap.
	Spatial Metric Group	Spatially filter candidates, then apply range or bound constraints.
	Metric Set Decision	Test a safety band or a margin-constrained closest-object decision.

Level-2: Compositional Depth Reasoning. Depth-VQA contains 1,503 questions in total: 757 Level 1 questions, 370 Level 2 Relative questions, and 376 Level 2 Metric questions. Level 2 uses grounded instances as reasoning operands, assigning each a stable identifier such as `obj_1`. Table 1 groups seven subbenches into the two tracks; Appendix Table 7 defines all question types. The Relative track uses exact-match accuracy: selections require the target identifier, and orderings require the full identifier sequence. Relative Overall Acc pools the 370 questions from the three reported subbenches. The Metric track uses accuracy for discrete answers and δ_1 for numerical gaps; Metric Overall Acc excludes δ_1 -scored questions. Appendix C provides question counts and scoring details.

5 EXPERIMENTAL SETUP

5.1 TRAINING PIPELINE

Training proceeds in two stages. Stage 1 freezes the vision-language backbone and trains the Level-0 decoder and high-resolution RGB branch for 80K steps. Stage 2 initializes from this checkpoint and jointly trains Level-1/Level-2 Depth-VQA and general VQA for 18K steps, with about 60K dense replay examples preserving Level-0 prediction. The dense-to-language interface is active for both instance levels. Because dense predictions and decoder features are detached before geometry-token construction, QA losses update the LoRA-adapted language modules and interface, while dense replay updates the decoder. We use Qwen3.5-4B (Qwen Team, 2026a); Appendix B provides data composition, preprocessing, optimization, and runtime settings.

5.2 EVALUATION SETUP

For Level 0, we evaluate metric depth on four outdoor datasets (Argoverse 2 (Wilson et al., 2021), Waymo (Sun et al., 2020), nuScenes (Caesar et al., 2020), and DDAD (Guizilini et al., 2020)) and five indoor datasets (Taskonomy (Zamir et al., 2018), ScanNet++ (Yeshwanth et al., 2023), HyperSim (Roberts et al., 2021), SUN RGB-D (Song et al., 2015), and Booster (Ramirez et al., 2022)). Predictions use ground-truth intrinsics and are scored in metric scale without post-hoc alignment; δ_1 is the primary metric, and the overall score averages the nine dataset-level results.

Depth-VQA follows the tasks and metrics defined in Section 4. Level-0 baselines include specialized metric-depth models and depth-aware VLMs; Depth-VQA baselines include general-purpose and depth-aware VLMs. All models use the same prompts and answer protocol. A matched DepthEvidence variant disables the dense-to-language interface at both instance levels to isolate geometry-token conditioning. Appendix C provides the complete baselines, sampling, and scoring protocols.

6 MAIN RESULTS

6.1 LEVEL-0 DENSE METRIC DEPTH PREDICTION

Because general-purpose VLMs lack dense metric-depth outputs, Table 2 compares DepthEvidence with depth-aware VLMs and specialized estimators. DepthEvidence achieves the highest nine-dataset average δ_1 of 0.7972, slightly above UniDepth V2 (0.7945) and 0.1033 above the strongest depth-aware baseline, DepthLM-12B. It exceeds DepthLM on eight datasets and leads on SUN RGB-D (0.8420). Its outdoor and indoor averages of 0.7970 and 0.7974 indicate balanced aggregate performance, although specialized models remain stronger on several individual datasets. Figure 3 complements Table 2 with two outdoor and two indoor examples; additional visualizations for the remaining five datasets are provided in Appendix Figure 5. DepthEvidence preserves fine structures and depth discontinuities more consistently than DepthVLM-4B and Youtu-VL-4B-Instruct while

Table 2: Level-0 dense metric depth comparison across outdoor and indoor benchmarks. We report $\delta_1 \uparrow$ and its unweighted average over all nine datasets. The best and second-best results in each column are shown in bold and underlined, respectively. Rankings are determined using the unrounded values. † DepthLM-12B follows its original sparse point-query protocol (Cai et al., 2026), using 8,192 randomly sampled pixels per dataset.

Method	Outdoor				Indoor				Avg.
	Argoverse2	Waymo	NuScenes	DDAD	Taskonomy	ScanNet++	HyperSim	SUN RGB-D	Booster
<i>VLMs Trained on Metric Depth Estimation</i>									
Youtu-VL-4B-Instruct	0.6290	0.5121	0.6443	0.2995	0.7561	0.6006	0.4285	0.3525	0.3471
DepthVLM-4B	0.7696	0.8056	0.8433	0.8232	0.6992	0.8568	0.4536	0.5089	0.2452
DepthLM-12B †	0.7768	0.7758	0.8310	0.6638	0.8218	0.7962	0.4358	0.5508	0.5934
<i>Specialized Vision Models</i>									
Depth Anything V2 outdoor	0.1304	0.7122	0.1701	0.1446	0.0008	0.0002	0.0415	0.0000	0.0000
Depth Anything V1 outdoor	0.2182	0.6564	0.3338	0.1779	0.0026	0.0001	0.0743	0.0001	0.0000
Depth Anything V1 indoor	0.0018	0.0019	0.0009	0.0015	0.4706	0.7604	0.2504	0.6427	0.5225
Depth Anything V2 indoor	0.1343	0.0188	0.1773	0.2891	0.2397	0.2322	0.8652	0.1651	0.6007
Metric3D	0.7801	0.8287	0.7691	0.8136	0.9467	0.0734	0.4745	0.0582	0.0434
Metric3D v2	<u>0.8589</u>	0.8774	<u>0.8555</u>	0.8700	<u>0.9389</u>	0.8809	0.7311	<u>0.6980</u>	0.1374
MetricAnything	0.5920	<u>0.8758</u>	0.8067	0.8736	0.8794	0.8966	0.8440	0.5713	0.5943
Depth Anything V3 metric-large	0.8051	0.8483	0.7996	0.7825	0.9189	0.8906	0.5800	0.6141	0.7175
UniDepth V2	0.8619	0.8502	0.8811	<u>0.8719</u>	0.8629	0.8602	<u>0.8648</u>	0.3969	<u>0.7004</u>
DepthEvidence-4B (Ours)	0.7948	0.8266	0.8122	0.7543	0.9072	0.8197	0.8213	0.8420	0.5970

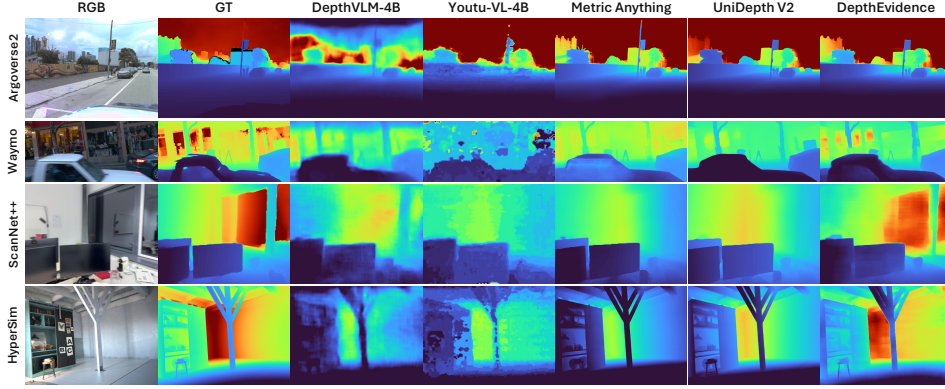


Figure 3: Qualitative dense depth on Argoverse2, Waymo, ScanNet++, and HyperSim. Columns show RGB, ground truth (GT), DepthVLM-4B, Youtu-VL-4B-Instruct, MetricAnything, UniDepth V2, and DepthEvidence. Outdoor GT is densified only for visualization; Table 2 uses the original sparse sensor GT. Each row shares one color scale; Figure 5 shows the other five datasets.

retaining the global layout. MetricAnything and UniDepth V2 sometimes produce smoother planes, whereas DepthEvidence remains competitive in boundary localization across both domains.

6.2 INSTANCE-LEVEL DEPTH ESTIMATION AND REASONING

Table 3: Instance-level depth estimation and reasoning results. The interface is enabled or disabled at both instance levels. Relative Overall Acc is computed over 370 questions, and Metric Overall Acc includes the 234 questions scored by accuracy. Bold and underlined entries denote the best and second-best results in each column.

Method	Level-1: Instance Metric Depth			Level-2: Compositional Depth Reasoning									
				Relative			Metric						
	$\delta_1 \uparrow$	RMSE $\downarrow$	AbsRel $\downarrow$	Spatial Pair	Spatial Group	Ordering	Overall Acc	Metric Pair Acc	Metric Pair $\delta_1 \uparrow$	Spatial Pair $\delta_1 \uparrow$	Spatial Group	Set Decision	Overall Acc
General-Purpose VLMs													
Qwen3.5-4B	0.0159	206.1221	9.5185	0.0930	0.0568	0.0102	0.0405	0.6304	0.2500	0.2234	0.3085	0.5532	0.4701
Gemini-3.1-Pro-Preview	0.4108	52.4680	2.0927	0.6395	0.5795	0.3724	0.4838	0.6304	0.2292	0.2340	0.3085	0.5957	0.4872
Qwen3.7-plus	0.0766	107.2556	3.0477	0.5814	0.6136	0.3827	0.4838	0.6522	0.2917	0.1489	0.3511	0.4681	0.4573
Seed-2.1-Turbo	0.2048	107.9360	2.7470	0.6977	0.6250	0.4133	0.5297	0.6304	0.1042	0.1596	0.3617	0.4362	0.4444
VLMs Trained on Metric Depth Estimation													
DepthLM-12B	0.7781	10.7460	0.1839	0.0000	0.0000	0.0000	0.0000	0.0000	0.1458	0.0532	0.0000	0.0000	0.0000
Youtu-VL-4B-Instruct	0.0264	22.1562	1.0852	0.1744	0.2955	0.1224	0.1757	0.2826	0.0417	0.1064	0.1915	0.3404	0.2692
DepthVLM-4B	0.6592	11.9164	0.2613	0.5000	0.3295	0.1837	0.2919	0.5870	0.0833	0.0957	0.3404	0.5213	0.4615
Agent Pipeline													
MetricAnything + Qwen3.5-4B	0.7318	11.9224	0.2363	0.3372	0.6136	0.4184	0.4459	0.5870	0.2917	0.2340	0.4149	0.6489	0.5427
MetricAnything + Qwen3.7-plus	0.7543	14.2529	0.2407	0.3140	0.7273	0.4541	0.4865	0.8478	0.4167	0.2979	0.4468	0.6596	0.6111
DepthEvidence w/o interface	0.8653	11.2159	0.2351	0.7442	0.5227	0.5408	0.5838	0.8261	0.4375	0.3511	0.7447	0.7660	0.7692
DepthEvidence (Ours)	0.8666	8.6062	0.1629	0.7907	0.5568	0.5867	0.6270	0.9130	0.5833	0.4043	0.8191	0.8511	0.8504

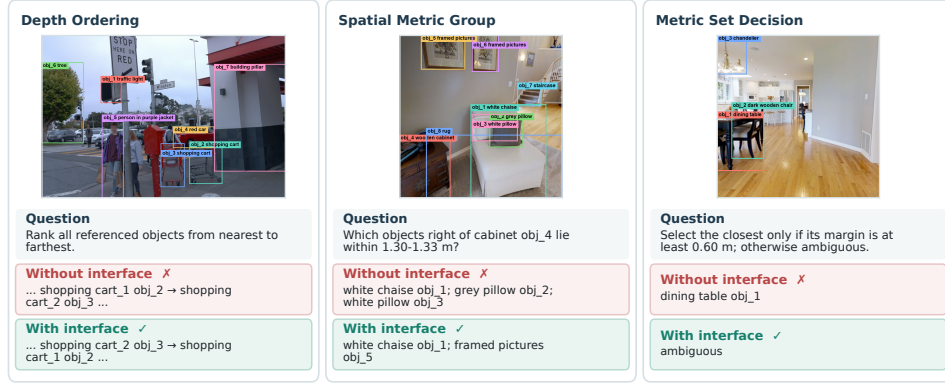


Figure 4: Representative Level-2 cases corrected by the dense-to-language interface. The three panels test complete depth ordering, spatially filtered metric-range selection, and margin-aware ambiguity decisions, respectively. Each panel contrasts matched variants without and with the interface; green answers match the ground truth. Aggregate results are reported in Table 3.

Table 4: General and spatial capability evaluation. Metrics follow their native scales; all are higher-is-better except Ego3D RMSE. Bold marks the best result, and “–” denotes unavailable results.

Method	STEM				General VQA			
	MathVision	MathVista	MMMU	MMMU-Pro	RealWorldQA	MMStar	MMBench	HallusionBench
Youtu-VL-4B-Instruct	–	76.50	61.10	43.00	74.60	71.10	83.90	59.10
GLM-4.1V-9B-Thinking	54.40	80.70	68.00	57.10	–	72.90	85.80	63.20
Qwen3.5-4B	74.60	85.10	77.60	66.30	79.50	78.30	89.40	65.00
DepthEvidence (Ours)	74.10	84.90	78.71	64.05	78.06	78.30	89.52	68.21
Method	Text Recognition and Document Understanding				Spatial Intelligence			
	AI2D	CC-OCR	OCRBench	CharXiv (RQ)	ERQA	EmbSpatialBench	Ego3D Acc.	Ego3D RMSE↓
Youtu-VL-4B-Instruct	85.60	–	81.30	43.80	–	–	45.63	19.27
GLM-4.1V-9B-Thinking	87.90	–	84.20	–	45.80	–	56.93	12.79
Qwen3.5-4B	89.60	76.70	85.00	70.80	54.00	81.30	62.82	13.17
DepthEvidence (Ours)	90.67	77.34	84.60	69.00	57.25	83.14	67.99	6.83

Table 3 evaluates instance-level metric estimation and compositional reasoning. Among the depth-aware baselines, DepthLM-12B (Cai et al., 2026) performs best on Level 1 ($\delta_1 = 0.7781$), whereas DepthVLM-4B (Yu et al., 2026) performs best on both Level 2 tracks (0.2919 Relative and 0.4615 Metric Overall Acc). DepthEvidence leads all evaluated methods on Level 1, with $\delta_1 = 0.8666$, RMSE = 8.6062, and AbsRel = 0.1629. It also achieves 0.6270 Relative and 0.8504 Metric Overall Acc, exceeding Seed-2.1-Turbo and Gemini-3.1-Pro-Preview by 9.73 and 36.32 percentage points, respectively, and DepthVLM-4B by 33.51 and 38.89 points. DepthEvidence further obtains the highest reported numerical-gap δ_1 on Metric Pair (0.5833) and Spatial Metric Pair (0.4043).

6.3 EFFECT OF THE DENSE-TO-LANGUAGE INTERFACE

Matched ablations in Table 3 isolate the interface effect. On Level 1, δ_1 is nearly unchanged (0.8653→0.8666), while RMSE and AbsRel decrease from 11.2159 to 8.6062 and from 0.2351 to 0.1629, suggesting fewer large metric errors. On Level 2, Overall Acc increases by 4.32 points on Relative and 8.12 points on Metric. The largest categorical gain is on Metric Pair (+8.69 points), while numerical-gap δ_1 rises from 0.4375 to 0.5833 on Metric Pair and from 0.3511 to 0.4043 on Spatial Metric Pair. Although MetricAnything + Qwen3.7-plus leads Relative Spatial Group (0.7273), its Relative and Metric Overall scores (0.4865/0.6111) trail DepthEvidence by 14.05/23.93 points, and both agent pipelines remain below DepthEvidence without the interface (Appendix C.6). Overall, the agent pipelines lag the full DepthEvidence system, indicating that external estimator–VLM coupling is less effective than our integrated design. Figure 4 visualizes representative corrections in depth ordering, spatially filtered metric-range selection, and margin-aware ambiguity decisions.

6.4 GENERAL VQA RETENTION AND SPATIAL UNDERSTANDING

Table 4 shows that DepthEvidence improves spatial reasoning while largely retaining the general capabilities of its Qwen3.5-4B initialization. Non-spatial scores change modestly: MMMU, MMBench, HallusionBench, AI2D, and CC-OCR improve, while the largest regression is 2.25 points on MMMU-Pro. All four spatial metrics improve: ERQA rises from 54.00 to 57.25, EmbSpatialBench

Table 5: Dense-decoder ablation on outdoor and indoor datasets ($\delta_1 \uparrow$).

SSI	HR branch	Outdoor		Indoor	
		Argoverse2	Waymo	Taskonomy	ScanNet++
–	–	0.7313	0.7505	0.7852	0.7349
✓	–	0.7547	0.7736	0.8175	0.7511
–	✓	0.7824	0.8129	0.8804	0.8081
✓	✓	0.7948	0.8266	0.9072	0.8197

Table 6: Geometry-interface ablation on Level-2 Relative and Metric Overall Acc.

Variant	$\mathcal{L}_{\text{ans}}$	$\mathcal{L}_{\text{raw}}$	$\mathcal{L}_{\text{ctx}}$	$\mathcal{L}_{\text{rank}}$	Relative Overall	Metric Overall
w/o interface	–	–	–	–	0.5838	0.7692
	✓	–	–	–	0.5838	0.7735
Interface	✓	✓	–	–	0.5946	0.8120
	✓	✓	✓	–	0.6135	0.8419
	✓	✓	✓	✓	0.6270	0.8504

from 81.30 to 83.14, and Ego3D accuracy from 62.82 to 67.99, while RMSE falls from 13.17 to 6.83 (−48.1%). DepthEvidence also outperforms Youtu-VL-4B-Instruct (Wei et al., 2026) and GLM-4.1V-9B-Thinking (GLM-V Team, 2025) on their commonly reported higher-is-better metrics and Ego3D RMSE.

7 ANALYSIS AND DISCUSSION

7.1 DENSE DECODER ABLATION

Table 5 shows that both the SSI objective and high-resolution branch increase δ_1 on each evaluated dataset, with their combination obtaining the highest values among the ablated variants. The HR branch has a larger measured effect indoors: used alone, it raises mean δ_1 by 0.0842 on the two indoor datasets versus 0.0568 outdoors. One possible explanation is the sparse evaluation coverage of outdoor datasets: their ground truth contains fewer evaluated pixels near object boundaries, so improvements in boundary prediction contribute less to the measured δ_1 than under denser indoor annotations.

7.2 GEOMETRIC SUPERVISION ABLATION

The interface is used at both instance levels; Table 6 focuses on its supervision design using the two Level-2 accuracy aggregates. Training the interface with only $\mathcal{L}_{\text{ans}}$ leaves Relative Overall Acc unchanged at 0.5838 and raises Metric Overall Acc from 0.7692 to 0.7735. Adding $\mathcal{L}_{\text{raw}}$ improves the two aggregates by 1.08 and 3.85 percentage points, respectively; $\mathcal{L}_{\text{ctx}}$ adds a further 1.89 and 2.99 points, and $\mathcal{L}_{\text{rank}}$ adds 1.35 and 0.85 points. The complete objective reaches 0.6270 Relative Overall Acc and 0.8504 Metric Overall Acc, improving over the model without the interface by 4.32 and 8.12 percentage points. These results support the utility of auxiliary geometric supervision, with a larger measured gain on the accuracy-based Metric tasks. A separate no-interface control with direct metric supervision remains clearly below the full interface, indicating that the main performance gain arises from the interface itself rather than numerical supervision alone; Appendix D.1 provides the detailed disentanglement analysis.

7.3 LIMITATIONS

The current setting assumes access to camera intrinsics and explicit object references. Its geometric evidence is therefore sensitive to errors in calibration, object grounding, and dense-depth prediction, particularly when depth ranges or decision margins are narrow. In addition, although cached continuation avoids repeated prefix computation, conditioning on geometry tokens incurs additional latency and memory relative to inference without the interface (Appendix E).

8 CONCLUSION

DepthEvidence connects dense metric depth prediction, instance-level metric depth estimation, and compositional depth reasoning within a shared MLLM. Its dense-to-language interface supplies object-aligned continuous geometry tokens, with geometric supervision encouraging metric-depth recoverability during language-context interaction. Experiments show competitive dense prediction and gains in object measurement, numerical depth gaps, and metric-constrained decisions. General VQA remains broadly stable relative to the base model, while spatial understanding improves on the evaluated benchmarks. These findings support dense prediction as both a supervised output and internal evidence for language reasoning. Future work can assess robustness to noisier camera intrinsics and object references, reduce inference cost as the object count grows, and extend the interface to other geometric evidence.

AI USE STATEMENT

We used Qwen3.7-plus and Seed-2.1-Turbo to propose object annotations and to generate and cross-verify synthetic Depth-VQA questions and answers, as described in Section 4 and Appendix C.1. We additionally used generative AI tools to improve manuscript readability. All benchmark examples were manually verified, and the authors reviewed all AI-assisted outputs and take responsibility for the final content.

REPRODUCIBILITY STATEMENT

Section 3 and Appendix A describe the model architecture and training objectives, while Section 5 and Appendix B provide the training pipeline, data composition, preprocessing, and optimization settings. Section 4 and Appendix C document benchmark construction, data partitioning, evaluation protocols, and metric definitions.

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A METHOD AND IMPLEMENTATION DETAILS

A.1 DENSE DEPTH BRANCH

A.1.1 DECODER AND CAMERA CONDITIONING

Given an RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, we retain three intermediate feature maps $\{\mathbf{V}_k\}_{k=1}^3$ from the vision encoder and reshape the final VLM hidden states at image-token positions into a spatial map $\mathbf{M}$. Following DPT (Ranftl et al., 2021), we normalize and project these four maps to $\{256, 512, 1024, 1024\}$ channels, resample them with strides $\{8, 4, 2, 1\}$, and map each level to a common width of 512 channels. The camera condition is applied to all four levels before fusion, while RGB features are injected into the three higher-resolution levels. Residual fusion blocks propagate the coarsest representation upward and merge it with progressively finer conditioned features. The resulting full-resolution feature is used both to predict dense metric depth and to construct object-level geometry representations.

Let $\kappa = (f_x, f_y, c_x, c_y)$ denote the supplied camera intrinsics for an image of original size $W_0 \times H_0$. We convert them to image-relative coordinates $\bar{\mathbf{k}} = [\bar{f}_x, \bar{f}_y, \bar{c}_x, \bar{c}_y] =$

$[f_x/W_0, f_y/H_0, c_x/W_0, c_y/H_0]$, and define $\phi_x = 2 \arctan(1/(2\bar{f}_x))$, $\phi_y = 2 \arctan(1/(2\bar{f}_y))$, and $r = W_0/H_0$. The decoder condition is the nine-dimensional vector

$$\mathbf{s}(\boldsymbol{\kappa}) = [\bar{\mathbf{k}}, \log \bar{f}_x, \log \bar{f}_y, \phi_x, \phi_y, r]. \quad (4)$$

An MLP encodes $\mathbf{s}(\boldsymbol{\kappa})$ as $\mathbf{c}$ and modulates each decoder level through $\text{FiLM}(\mathbf{F}_k, \mathbf{c}) = \mathbf{F}_k \odot (1 + \gamma_k(\mathbf{c})) + \beta_k(\mathbf{c})$.

A.1.2 HIGH-RESOLUTION RGB REFINEMENT

The RGB branch processes the input image with a convolutional stem followed by three stride-2 stages, producing feature maps $\{\mathbf{R}_k\}_{k=1}^3$ at strides $\{2, 4, 8\}$. After channel projection and spatial alignment, each RGB map is added to its corresponding DPT level through a learned residual spatial gate, $\tilde{\mathbf{F}}_k = \mathbf{F}_k + \mathbf{G}_k \odot \mathbf{R}_k$, where $\mathbf{G}_k = \text{clip}(b_k + 1.5 \tanh g_k([\mathbf{F}_k; \mathbf{R}_k]), 0, 3)$. The function g_k predicts a spatially varying residual from the concatenated DPT and RGB features, and the base gate b_k is initialized to 0.01. After the final fusion block, a convolution reduces the feature width, a second FiLM layer applies the camera condition, and the feature map is bilinearly resized to the target resolution. The metric prediction is $\hat{\mathbf{D}} = \text{clip}(\text{softplus}(h(\mathbf{F}))) + d_{\min}, d_{\min}, d_{\max})$, with $d_{\min} = 0.05$ m and $d_{\max} = 250$ m.

A.1.3 DENSE OBJECTIVE

The dense objective combines the absolute relative and scale-invariant log losses (Eigen et al., 2014) with structural supervision on inverse depth. Specifically, $\mathcal{L}_{\text{der}}$ and $\mathcal{L}_{\text{lap}}$ match Scharr derivatives and Laplacian responses at scales $\{1, 2, 4, 8\}$. For the edge-guided stable local scale-and-shift invariant term $\mathcal{L}_{\text{ssi}}$, we sample regions centered on strong RGB edges and transform depth using $q(D) = \log(1 + 50/(10^{-4} + D))$. Prediction and target values are robustly normalized and compared both within the sampled local regions and over the full image.

A.2 DEPTH-VQA INSTRUCTIONS AND GROUNDING

A Level-1 example contains an image $\mathbf{I}$ and a grounded target object o_i , and asks for instance-level metric depth estimation. The default target d_i is the mean of valid depths within the object’s instance mask; queries can also request center-point, median, nearest, or farthest depth. A Level-2 example contains grounded objects $\mathcal{O} = \{o_i\}_{i=1}^N$ and a question q requiring compositional depth reasoning in the Relative or Metric track. Relative questions ask for pairwise comparisons, closest or farthest object selection, or depth ordering. Metric questions ask for numerical mean-depth gaps, comparisons against a threshold, selection within a depth range or bounds, or decisions under a required depth margin. Compositional questions combine these operations with image-space relations to select the relevant objects. For a margin-based closest-object decision, the target is *ambiguous* when the closest object fails to meet the specified margin.

Depth convention and object statistics. Camera depth denotes the optical-axis coordinate Z of a point in the camera frame, measured in meters, rather than its Euclidean distance to the camera center. Object depth denotes the statistic requested for an instance. For Level 1, let Ω_i contain the pixels inside the ground-truth instance mask with valid camera depth $D(p)$. The mean target is $d_i^{\text{mean}} = |\Omega_i|^{-1} \sum_{p \in \Omega_i} D(p)$, and the median target is the median over the same set. The nearest and farthest targets are the 5th and 95th percentiles of valid mask depths, respectively; center-point depth is a point measurement rather than a region aggregate. The mask defines the target statistics, while a box or referring prompt identifies the object to the model. The interface pools predicted evidence over box regions (Appendix A.3.1). Mean camera-depth gaps compare optical-axis depth statistics, not Euclidean distances between objects in 3D.

A box prompt specifies the object category and bounding box, with coordinates normalized to a 0–1000 image coordinate system. A referring prompt identifies the object with a natural-language description enclosed by object-reference tokens. Each object in a Level-2 prompt receives a stable identifier such as `obj_1`, and the prompt lists its identifier, category, and grounding cue before the question.

Every Depth-VQA instruction embeds the camera intrinsics in the user prompt as `<focal> f_x f_y c_x c_y </focal>`. The token name `<focal>` is retained in the implementation, but the block

contains both focal lengths and principal-point coordinates. The language-modeling loss is applied only to the final answer: a metric depth value for Level 1, or a numerical depth gap, an object or set selection, an ordering, or a discrete comparison or constraint decision for Level 2, including *ambiguous* for an unmet depth-margin requirement. Input positions are excluded as language-modeling targets; auxiliary geometric losses supervise the geometry representations as specified in Appendix A.3.4.

A.3 DENSE-TO-LANGUAGE INTERFACE

A.3.1 CONSTRUCTION

The same dense-to-language interface is used for Level 1 single-object queries and Level 2 multi-object questions. The dense decoder produces a metric depth map $\hat{\mathbf{D}}$ and a full-resolution fused feature map $\mathbf{F}^d$. For each grounded object o_i , let $\mathbf{b}_i = [x_i^1, y_i^1, x_i^2, y_i^2]$ be its normalized box and let $\mathcal{R}_i$ denote the corresponding region. We average-pool $\mathbf{F}^d$ inside $\mathcal{R}_i$ to obtain an appearance-and-shape descriptor $\mathbf{r}_i$, and summarize the predicted depths in the same region as $\mathbf{s}_i = [\mu_i, \sigma_i, d_i^{\min}, d_i^{\max}, d_i^{\text{ctr}}, \log \mu_i, a_i, \rho_i]$. These entries denote the mean, standard deviation, minimum, maximum, center depth, log-mean depth, box area, and aspect ratio, respectively.

For a box-grounded object, we directly use the box given in the prompt. For a referring expression enclosed by `<|object_ref_start|>` and `<|object_ref_end|>`, we take the VLM hidden state at the closing reference token as an object query. A lightweight box-regression head, implemented as layer normalization followed by a two-layer MLP, predicts $\hat{\mathbf{b}}_i \in [0, 1]^4$. The predicted box is supervised by the ground-truth region during training and is used for differentiable region pooling and inference.

The projected object tokens are organized as

$$\mathcal{B}_G = [\mathbf{g}_{\text{start}}, e(\text{obj_1:}), \mathbf{g}_{1,1}, \mathbf{g}_{1,2}, \dots, e(\text{obj_N:}), \mathbf{g}_{N,1}, \mathbf{g}_{N,2}, \mathbf{g}_{\text{end}}], \quad (5)$$

where $e(\text{obj_i:})$ is the ordinary language-embedding sequence of an object anchor and $\mathbf{g}_{\text{start}}$ and $\mathbf{g}_{\text{end}}$ are learned boundary vectors. Placeholder `<geometry>` embeddings at the dynamic positions are replaced by the corresponding object vectors before the continuation enters the language model. The bank therefore contains $2N + 2$ dynamically generated vectors for N objects, in addition to the ordinary textual anchors. Its positions serve as conditioning context and are excluded as language-modeling targets. The answer loss nevertheless backpropagates through the geometry representations to the projector. The bank contains predicted geometry; ground-truth depth statistics serve as targets for the auxiliary representation objectives.

A.3.2 TOKEN CAPACITY: WHY TWO PER OBJECT?

We use $K = 2$ as a compact multi-token bottleneck for each grounded object. The fused descriptor $[\mathbf{r}_i; \mathbf{b}_i; \mathbf{s}_i]$ contains decoder features, image location, and several statistics of the predicted metric depth. Compressing all of this evidence into a single VLM vector would impose a narrower information bottleneck. Instead, the final layer of P_g produces a $2h$ -dimensional output that is reshaped into two independently parameterized vectors in $\mathbb{R}^h$. Both vectors are computed from the complete object descriptor; we do not prescribe one token for appearance and the other for metric depth. A shared object-index embedding binds the pair to the same instance, while distinct slot embeddings break their symmetry and allow the language model to use the two positions differently. The resulting specialization, if any, is therefore learned rather than hand-crafted. This added representational bandwidth requires only one additional dynamic position per object relative to $K = 1$: for N grounded objects, the geometry bank contains $2N$ object-geometry positions instead of N . Thus, $K = 2$ provides a controlled capacity increase while keeping the interface sequence length proportional to the number of grounded objects; the subsequent attention cost follows the standard dependence on the resulting total sequence length.

A.3.3 CACHED GEOMETRY CONTINUATION

At both instance levels, let $\mathbf{P}$ contain the image, camera-intrinsics block, grounded object descriptions, and question. Processing this prefix once produces contextualized states $\mathbf{H}_P$ and a layer-wise cache

$\mathcal{C}_P$. The dense branch uses the visual positions in $\mathbf{H}_P$ together with the multi-level vision features to construct $\mathcal{B}_G$, after which the cache is extended and the answer is generated as

$$(\mathbf{H}_P, \mathcal{C}_P) = f_\theta(\mathbf{P}; \text{cache} = \text{on}), \quad \mathcal{C}_{P,G} = f_\theta(\mathcal{B}_G; \mathcal{C}_P), \quad \mathbf{Y} \sim f_\theta(\cdot; \mathcal{C}_{P,G}). \quad (6)$$

In a full-attention layer, $\mathcal{C}_P$ contains the prefix keys and values; queries from the geometry continuation attend to these cached tensors and to preceding continuation positions. The language backbone also contains linear-attention layers, which summarize the prefix with a causal-convolution state and a recurrent state rather than a conventional key-value sequence. We clone these differentiable prefix states and use them to initialize the chunk kernels that process the multi-token geometry continuation. Both attention types therefore process the continuation as if $[\mathbf{P}; \mathcal{B}_G; \mathbf{Y}]$ had been evaluated as one causal sequence. Caching changes the computation schedule without changing causal order or exposing the geometry tokens as visible outputs.

A.3.4 REPRESENTATION OBJECTIVES

For object i , let $\bar{\mathbf{g}}_i = (\mathbf{g}_{i,1} + \mathbf{g}_{i,2})/2$ be the raw representation before it enters the language model, and let $\bar{\mathbf{h}}_i = (\mathbf{h}_{i,1} + \mathbf{h}_{i,2})/2$ be the final-layer representation at the same geometry positions after cached continuation. Let $\mathbf{d}_i$ contain five ground-truth object-depth statistics: center, mean, median, nearest, and farthest depth. Two lightweight heads predict their logarithms from the raw and contextualized representations:

$$\hat{\mathbf{z}}_i^{\text{raw}} = H_{\text{raw}}(\bar{\mathbf{g}}_i), \quad \hat{\mathbf{z}}_i^{\text{ctx}} = H_{\text{ctx}}(\bar{\mathbf{h}}_i), \quad \mathbf{z}_i = \log \mathbf{d}_i. \quad (7)$$

The valid-depth ratio w_i weights the corresponding smooth- ℓ_1 regressions,

$$\mathcal{L}_{\text{raw}} = \frac{\sum_i w_i \text{SmoothL1}(\hat{\mathbf{z}}_i^{\text{raw}}, \mathbf{z}_i)}{\sum_i w_i}, \quad \mathcal{L}_{\text{ctx}} = \frac{\sum_i w_i \text{SmoothL1}(\hat{\mathbf{z}}_i^{\text{ctx}}, \mathbf{z}_i)}{\sum_i w_i}. \quad (8)$$

At both instance levels, we operationalize metric information preservation through the recoverability of these ground-truth statistics from both representations. The raw and contextual losses supervise the projected tokens and their states after language-context interaction against the same targets, encouraging metric content to remain accessible to a prediction head. For multi-object examples, we additionally supervise relative depth using

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \text{softplus}(-s_{ij}(\hat{z}_i - \hat{z}_j)), \quad s_{ij} = \text{sign}(\log d_i^{\text{mean}} - \log d_j^{\text{mean}}), \quad (9)$$

where $\hat{z}_i$ is the contextual prediction of mean log-depth and $\mathcal{P}$ contains object pairs whose ground-truth log-depth gap exceeds a small ambiguity margin. The differentiable cache allows the answer and contextual geometry objectives to update the continuation adapters and trainable prefix-pathway parameters. The geometry projector receives gradients from the answer, raw-depth, contextual-depth, and ranking losses. We detach $\hat{\mathbf{D}}$ and $\mathbf{F}^d$ before object projection, so these objectives do not perturb the pretrained dense decoder, which continues to be optimized by $\mathcal{L}_{\text{dense}}$ on dense-prediction batches.

B DATA AND TRAINING DETAILS

B.1 DATA COMPOSITION

The Level-0 corpus combines approximately 6.7M images from LingBot-Depth, Taskonomy, Waymo, DDAD, nuScenes, Argoverse 2, and ScanNet++. We apply depth completion to outdoor sources with sparse sensor depth so that every source image is paired with a dense depth map, and preserve the camera intrinsics associated with each image. The dense branch converts these intrinsics into its numerical camera condition, whereas Depth-VQA prompts provide the raw (f_x, f_y, c_x, c_y) values in a textual camera-intrinsics block. All dense-benchmark frames are excluded from training. The instance-level corpus contains approximately 1.5M Level-1 and Level-2 examples sampled from LingBot-Depth, Waymo, DDAD, Argoverse 2, nuScenes, and Taskonomy. Stage 2 combines these examples with roughly 771K general visual-instruction examples and about 60K Level-0 replay samples, with stream frequencies proportional to corpus size.

B.2 OPTIMIZATION AND RUNTIME

Optimization. We use Qwen3.5-4B as the vision-language backbone and AdamW with weight decay 0.01 and a 3% warmup ratio. During Stage 1, the vision encoder and language model are frozen, and the dense branch is trained with a learning rate of 10^{-3} . During Stage 2, decoder-side modules and LoRA-adapted language parameters use a learning rate of 10^{-4} , while the vision encoder uses 10^{-7} . LoRA is applied to attention and feed-forward projections with rank 32, scaling factor 64, and dropout 0.05.

Runtime settings. Dense inputs are resized to 448×448 . Instruction-tuning sequences contain at most 4,096 tokens, including up to 2,048 visual tokens. We train in bfloat16 with FlashAttention-2 and an effective global batch size of 256 for both dense and VQA updates. Stage 2 runs on four nodes with 32 NVIDIA A100 GPUs in total.

C BENCHMARK CONSTRUCTION AND EVALUATION PROTOCOLS

C.1 BENCHMARK CONSTRUCTION AND QUALITY CONTROL

Dense depth preparation. Outdoor datasets provide sparse sensor measurements, so we first apply depth completion to obtain a dense depth map for every source image. These image–depth pairs form the Level-0 pool and provide the metric field from which instance-level targets are derived.

Level-1 instance construction. For each image, Qwen3.7-plus (Qwen Team, 2026b) and Seed-2.1-Turbo (ByteDance Seed Team, 2026) independently identify the major objects and propose a category and grounding box for each one. We pool their candidates and discard boxes covering less than 1% of the image or lying too close to image boundaries. The remaining category–box pairs prompt SAM 3 (Carion et al., 2026) to produce instance masks, which pass a second mask-quality filter. For each retained mask, we combine its label, box, and dense depth values to compute center-point, mean, median, nearest, and farthest targets; the latter two are the 5th and 95th percentiles of valid mask depths, respectively. The resulting grounded instance records constitute the Level-1 data. Using two proposal models reduces dependence on model-specific object and category preferences during automatic construction.

Level-2 question construction. We sample multiple Level-1 instance records from an image according to the Relative and Metric task taxonomy in Table 7. Given the selected object labels, boxes, and depth statistics, either Qwen3.7-plus or Seed-2.1-Turbo generates a candidate question and answer. The other model then checks question validity and answer correctness, and invalid examples are removed. This cross-model generation–verification protocol reduces reliance on the phrasing and reasoning patterns of a single generator.

Human review and distribution control. We manually inspect 1% of the training samples across Levels 0–2, aggregate recurring failure cases, and translate them into targeted filters applied to the training corpus. Every benchmark example undergoes manual verification. For video data, we favor different objects when constructing Level-1 and Level-2 questions within the same scene and limit reuse of the same instance across task types, reducing repeated-instance bias.

C.2 BENCHMARK COMPOSITION AND PARTITIONING

Table 7: Detailed composition of the Relative Depth and Metric Depth tracks in the Level-2 benchmark. “Subset” orders a specified subset of objects, whereas “all” orders every referenced object.

Track	Subbench	Raw question type	Required reasoning
Relative Depth	Spatial Pair	closer_farther	Decide which of two objects is closer to or farther from the camera.
		front_behind	Decide which of two objects is in front of or behind the other.
		spatial_depth	Resolve a left/right reference, then compare the selected object with another object in depth.
	Spatial Group	spatial_depth_group	Filter objects by a spatial relation, then select the nearest or farthest member.
	Depth Ordering	depth_ordering_closest depth_ordering_farthest depth_ordering_subset depth_ordering_all	Select the closest object from all candidates. Select the farthest object from all candidates. Order a specified subset of objects by depth. Order all referenced objects by depth.
Metric Depth	Metric Pair	metric_depth_gap	Compute the numerical difference in mean camera depth between two objects.
		metric_gap_threshold	Determine whether the mean-depth gap between two objects reaches a specified metric threshold.
	Spatial Metric Pair	spatial_metric_gap	Resolve a 2D spatial reference, then compute the absolute mean-depth difference to another object.
	Spatial Metric Group	spatial_range_filter	Apply a spatial filter, then select objects whose mean depths fall inside a specified metric range.
		compositional_metric_constraint	Combine a spatial condition, a metric lower bound, and an object-referenced upper-depth bound.
	Metric Set Decision	camera_safety_band	Determine whether any referenced object lies within a specified mean-depth range from the camera.
		metric_margin_decision	Select the closest object only when its depth advantage exceeds a metric margin; otherwise answer <i>ambiguous</i> .

Benchmark partitioning. For datasets with official partitions, benchmark examples are sampled exclusively from their official validation or test splits. For datasets without an official evaluation partition, we randomly hold out complete scenes using a fixed seed and sample benchmark examples only from these held-out scenes. No held-out scene appears in either the dense-depth or instruction-tuning corpora.

Table 8: Compact statistics for the dense and instance-level depth benchmarks. Depth-VQA contains 1,503 questions in total.

Component	Sampling unit	Scale or coverage	Task composition
Level-0 dense depth	Image	4,228 images from nine datasets	500 images from each of eight datasets and all 228 Booster images
Level-1 instance depth	Grounded-object query	757 queries from seven datasets	Center, nearest, farthest, mean, and median metric depth; box or referring prompts
Level-2 Relative	Multi-object question	370 questions	Spatial Pair (86), Spatial Group (88), and Depth Ordering (196)
Level-2 Metric	Multi-object question	376 questions	Metric Pair (46 threshold + 48 numerical gap), Spatial Metric Pair (94), Spatial Metric Group (94), and Metric Set Decision (94)

C.3 LEVEL-0 DENSE-DEPTH EVALUATION

Sampling and ground truth. We select 500 frames from each of Argoverse 2, Waymo, nuScenes, DDAD, Taskonomy, ScanNet++, HyperSim, and SUN RGB-D using a fixed random seed, and use all 228 Booster images spanning 38 scenes and six exposure settings. All selected frame identities are removed from the dense training manifest. The automotive datasets are evaluated against their original sensor-derived metric ground truth rather than depth-completion outputs. Booster disparity is converted to metric depth using its official calibration and evaluated with the official validity mask.

Scoring. Ground-truth camera intrinsics are supplied to every method evaluated in Table 2. Predictions are compared directly in metric scale without median scaling or scale-and-shift alignment. For the first eight datasets, valid pixels must be finite and lie within $[0.05, 250]$ meters; Booster follows its official mask and $[0.001, 10]$ meter range. We compute AbsRel, RMSE, log RMSE, SLog, and δ_1 , δ_2 , and δ_3 over valid pixels. The main comparison reports δ_1 , with the cross-dataset score computed as the unweighted mean of the nine dataset-level results.

C.4 DEPTH-VQA EVALUATION

Level 1 metrics are computed over the requested instance-depth values after parsing the numerical answer in meters. For the Level 2 Relative track, a selection answer is correct only if it contains the target identifier, and an ordering answer is correct only if the full predicted identifier sequence matches the reference. Relative Overall Acc is computed over the 86 Spatial Pair, 88 Spatial Group, and 196 Depth Ordering questions (370 in total). In the Metric track, Metric Pair separates 46 `metric_gap_threshold` questions scored by accuracy from 48 `metric_depth_gap` questions scored by δ_1 . Spatial Metric Pair reports δ_1 on its 94 `spatial_metric_gap` questions. Metric Overall Acc divides the total correct answers on the threshold, Spatial Metric Group, and Metric Set Decision questions by $46 + 94 + 94 = 234$. The 142 numerical gap questions are excluded from Overall Acc and evaluated in their respective δ_1 columns.

C.5 GENERAL-CAPABILITY BENCHMARKS AND BASELINES

Benchmarks. Table 4 covers MathVision (Wang et al., 2024a), MathVista (Lu et al., 2024), MMMU (Yue et al., 2024a), MMMU-Pro (Yue et al., 2024b), RealWorldQA (xAI, 2024), MMStar (Chen et al., 2024b), MMBench (Liu et al., 2024), HallusionBench (Guan et al., 2024), AI2D (Kembhavi et al., 2016), CC-OCR (Yang et al., 2025), OCRBench (Liu et al., 2023), CharXiv (Wang et al., 2024b), ERQA (Google DeepMind, 2025), EmbSpatialBench (Du et al., 2024), and Ego3D-Bench (Gholami et al., 2026).

Baselines. For Level-0 dense prediction, the specialized vision models are Metric3D (Yin et al., 2023), Metric3D v2 (Hu et al., 2024), UniDepth V2 (Piccinelli et al., 2026), Depth Anything V1–V3 (Yang et al., 2024a;b; Lin et al., 2026), and MetricAnything (Ma et al., 2026); the depth-aware VLMs are DepthLM (Cai et al., 2026) and DepthVLM (Yu et al., 2026). For instance-level Depth-VQA, the general-purpose VLMs are Qwen3.5-4B (Qwen Team, 2026a), Gemini-3.1-Pro-Preview (Google DeepMind, 2026), Qwen3.7-plus (Qwen Team, 2026b), and Seed-2.1-Turbo (ByteDance Seed Team, 2026), while the metric-depth VLMs are DepthLM-12B, DepthVLM-4B, and Youtu-VL-4B-Instruct.

C.6 AGENT-PIPELINE BASELINE IMPLEMENTATION

The agent baselines in Table 3 use a fixed tool-augmented inference pipeline. For each benchmark question, MetricAnything receives the original RGB image and the camera intrinsics provided by the benchmark and predicts a pixel-level metric depth map in meters. We then crop this prediction with each benchmark-provided grounding box and compute a structured set of regional depth statistics for the corresponding identifier, such as `obj_i`. The resulting object-indexed depth table is supplied together with the original image and question to either Qwen3.5-4B or Qwen3.7-plus. The reasoner returns its final response between `<answer>` and `</answer>`, and the parsed response is evaluated by the original benchmark scorer. The pipeline therefore exposes predicted metric measurements through an explicit textual table while preserving the benchmark questions, object grounding, and scoring protocol; benchmark boxes are used only to extract regions from the predicted depth map, and no ground-truth depth values are given to the reasoner.

D ADDITIONAL INTERFACE ABLATION

D.1 DISENTANGLING INTERFACE AND METRIC SUPERVISION

Table 9 tests whether the gain from the dense-to-language interface can be explained by additional supervision on metric values alone. The new *No interface + metric supervision* control removes the interface and adds auxiliary language targets for every instance referenced by a QA example. These targets ask the model to predict the instance’s center, mean, median, nearest, and farthest depths (e.g., “Given `obj_1`, predict its center depth”) and are optimized with the language loss. The remaining three rows reproduce the corresponding configurations from Table 6.

Table 9: Disentangling direct metric supervision from the dense-to-language interface on Level-2 Relative and Metric Overall Acc.

Variant	Relative Overall	Metric Overall
No interface	0.5838	0.7692
No interface + metric supervision	0.5865	0.7949
Interface + $\mathcal{L}_{\text{ans}}$	0.5838	0.7735
Full interface	0.6270	0.8504

Without the interface, direct metric supervision improves Relative Overall Acc by 0.27 percentage points and Metric Overall Acc by 2.56 points. The larger Metric gain confirms that the auxiliary language targets provide useful numerical depth supervision. However, the full interface remains ahead of this control by 4.05 points on Relative Overall Acc and 5.56 points on Metric Overall Acc. Within this comparison, metric supervision therefore explains only part of the improvement: the full interface provides an additional benefit beyond exposing the language model to referenced-object depth values through auxiliary text targets.

E INFERENCE EFFICIENCY

We compare three inference modes with end-of-sequence stopping disabled and a fixed output length of 64 tokens. *No-interface* generates answers without geometry tokens. *Two-pass* first extracts geometry and then reprocesses the multimodal prefix with the geometry tokens to generate the answer. *Single-prefill* reuses the prefix cache and processes the geometry tokens as a continuation, as described in Appendix A.3.3. Table 10 reports generation latency, end-to-end latency, token throughput, and peak allocated GPU memory under this fixed-length setting.

Table 10: Inference efficiency with a fixed output length of 64 tokens and end-of-sequence stopping disabled. Single-prefill is the cached geometry-continuation scheme used by DepthEvidence.

Mode	Generation latency (s)↓	End-to-end latency (s)↓	Throughput (tokens/s)↑	Peak allocated memory (GiB)↓
No-interface	4.174	4.289	15.333	10.36
Two-pass	4.535	4.710	14.111	13.56
Single-prefill	4.323	4.491	14.804	13.73

Single-prefill reduces generation latency from 4.535 to 4.323 s and end-to-end latency from 4.710 to 4.491 s relative to two-pass inference, reductions of 4.67% and 4.65%, respectively. Throughput increases by 4.91%, from 14.111 to 14.804 tokens/s. Relative to no-interface inference, the end-to-end overhead is 0.202 s (4.71%). These results support reusing the multimodal prefix to limit the runtime cost of incorporating dense geometry into language reasoning. The benefit is in latency rather than memory: single-prefill uses 13.73 GiB of peak allocated memory, 0.17 GiB more than two-pass and 3.37 GiB more than no-interface. Thus, cached continuation reduces repeated prefix computation while retaining an additional memory cost for geometry-conditioned inference.

F ADDITIONAL QUALITATIVE RESULTS

F.1 DENSE METRIC DEPTH

Figure 5 extends the qualitative comparison in Section 6.1 to the five remaining dense benchmarks.

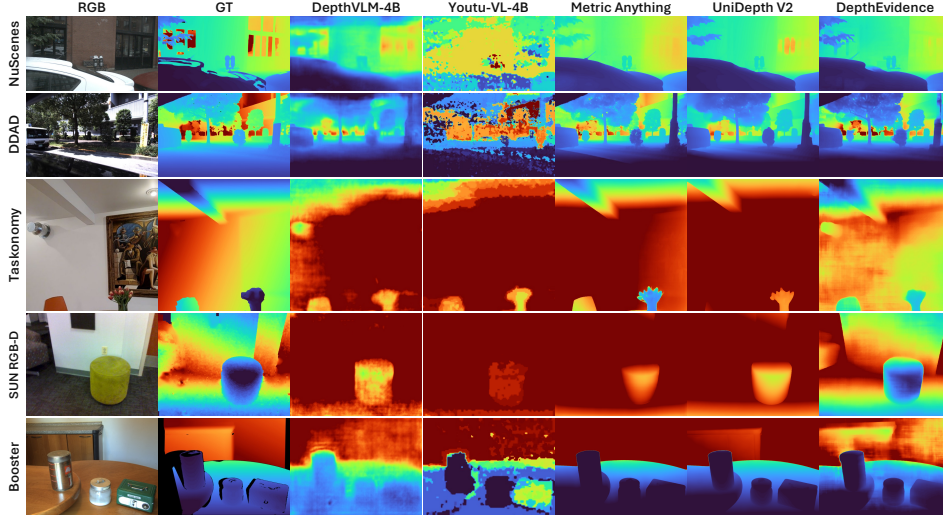


Figure 5: Additional qualitative dense metric depth comparisons on NuScenes, DDAD, Taskonomy, SUN RGB-D, and Booster. Columns show RGB, ground truth (GT), DepthVLM-4B, Youtu-VL-4B-Instruct, MetricAnything, UniDepth V2, and DepthEvidence. Outdoor GT is densely completed for visualization only; quantitative evaluation uses the original sparse sensor GT. All depth maps within a row use the same color scale.

F.2 INSTANCE-LEVEL DEPTH-VQA

We provide selected qualitative cases, including the input image, question, ground-truth answer, and the outputs of Gemini-3.1-Pro, DepthEvidence without the interface, and DepthEvidence. A green checkmark (✓) denotes a correct benchmark prediction, and a red cross (✗) denotes an incorrect one. Box-grounded questions are paired with the annotated box image, whereas referring-prompt questions retain the original image. Level 1 cases are grouped by dataset; Level 2 cases follow the subbench taxonomy of Table 1 and cover every Relative Depth and Metric Depth subbench. Examples reproduced in Figure 4 are omitted here, except for the sole selected Depth Ordering example.

F.2.1 LEVEL-1 CASES GROUPED BY DATASET

These examples span center-point, mean, median, nearest, and farthest instance-depth queries. The displayed target specification is the semantic reference seen by the model; box coordinates remain in the question for box-grounded cases.

LingBot-Depth



Mean depth; box prompt

Question. Given this image and the target object, estimate the average depth of the target region from the camera. Target: classroom sign with bbox_2d [670, 38, 888, 529]

Ground truth: 1.25 m

Gemini-3.1-Pro: 2.11 m ✗

DepthEvidence w/o Interface: 1.31 m ✓

DepthEvidence: 1.32 m ✓

Waymo



Farthest depth; box prompt

Question. Given this image and the target object, estimate the farthest depth of the target region from the camera. Target: silver sedan with bbox_2d [819, 454, 999, 562]

Ground truth: 27.167 m

Gemini-3.1-Pro: 41.11 m ✗

DepthEvidence w/o Interface: 26.62 m ✓

DepthEvidence: 27.3 m ✓



Median depth; referring prompt

Question. Given this image and the target object, estimate the median depth of the target region from the camera. Target: pole referred as "the tall, dark metal pole on the far left sidewalk near the traffic light"

Ground truth: 16 m

Gemini-3.1-Pro: 10.11 m ✗

DepthEvidence w/o Interface: 15 m ✓

DepthEvidence: 15.56 m ✓

DDAD



Mean depth; box prompt

Question. Given this image and the target object, estimate the average depth of the target region from the camera. Target: green trees with bbox_2d [736, 248, 999, 514]

Ground truth: 50.94 m

Gemini-3.1-Pro: 101.12 m ✗

DepthEvidence w/o Interface: 52.4 m ✓

DepthEvidence: 55.3 m ✓

Argoverse 2



Nearest depth; referring prompt

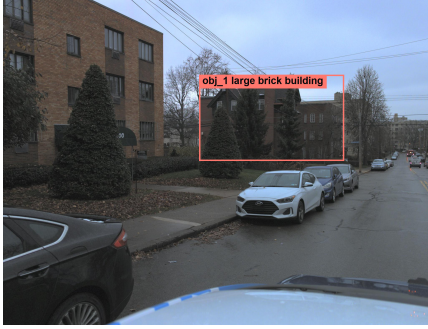
Question. Given this image and the target object, estimate the nearest depth of the target region from the camera. Target: white car hood referred as "the white car hood with 'ARGO SELF-DRIVING TEST VEHICLE' text at the bottom of the image"

Ground truth: 1.15 m

Gemini-3.1-Pro: 1.8 m ✗

DepthEvidence w/o Interface: 1.06 m ✓

DepthEvidence: 1.03 m ✓



Nearest depth; box prompt

Question. Given this image and the target object, estimate the nearest depth of the target region from the camera. Target: large brick building with bbox_2d [458, 230, 797, 495]

Ground truth: 37.46 m

Gemini-3.1-Pro: 10.82 m ✗

DepthEvidence w/o Interface: 40.5 m ✓

DepthEvidence: 36.5 m ✓

nuScenes



Median depth; box prompt

Question. Given this image and the target object, estimate the median depth of the target region from the camera. Target: road with bbox_2d [2, 530, 999, 998]

Ground truth: 7.19 m

Gemini-3.1-Pro: 14.11 m ✗

DepthEvidence w/o Interface: 7.53 m ✓

DepthEvidence: 7.06 m ✓

Taskonomy



Mean depth; box prompt

Question. Given this image and the target object, estimate the average depth of the target region from the camera. Target: blue bowl with bbox_2d [734, 713, 992, 828]

Ground truth: 1.53 m

Gemini-3.1-Pro: 1.81 m ✓

DepthEvidence w/o Interface: 1.4 m ✓

DepthEvidence: 1.48 m ✓

F.2.2 LEVEL-2 CASES GROUPED BY TABLE 1 SUBBENCH

The cases below retain the same Relative Depth and Metric Depth organization used by the benchmark. Annotated boxes identify the candidate objects and their stable `obj_i` identifiers.

Relative Depth

Spatial Pair.



spatial_depth; box prompt

Question. Which is closer to the camera, the object to the left of silver pickup truck obj_2 or sidewalk obj_3?

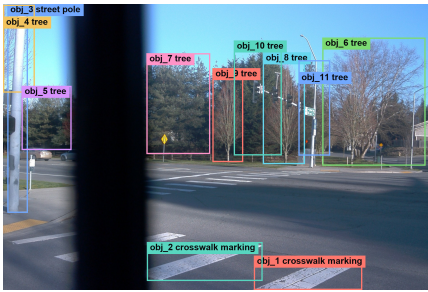
Ground truth: red sedan obj_1.

Gemini-3.1-Pro: obj_3 ✗

DepthEvidence w/o Interface: red sedan obj_1. ✓

DepthEvidence: red sedan obj_1. ✓

Spatial Group.



spatial_depth_group; box prompt

Question. Among the objects to the left of tree_3 obj_6, which is closest to the camera?

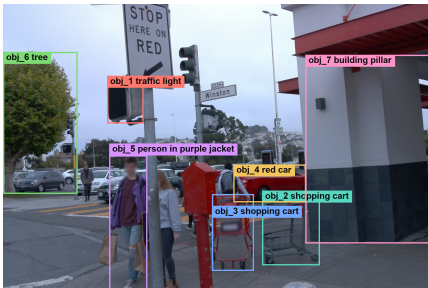
Ground truth: crosswalk marking_1 obj_1.

Gemini-3.1-Pro: obj_3 ✗

DepthEvidence w/o Interface: crosswalk marking_1 obj_1. ✓

DepthEvidence: crosswalk marking_1 obj_1. ✓

Depth Ordering.



depth_ordering_all; box prompt

Question. Rank the main objects in this scene by their distance to the camera, starting from the closest.

Ground truth: traffic light obj_1, person in purple jacket obj_5, shopping cart_2 obj_3, shopping cart_1 obj_2, building pillar obj_7, red car obj_4, tree obj_6.

Gemini-3.1-Pro: obj_1, obj_5, obj_7, obj_3, obj_2, obj_4, obj_6 ✗

DepthEvidence w/o Interface: traffic light obj_1, person in purple jacket obj_5, shopping cart_1 obj_2, shopping cart_2 obj_3, building pillar obj_7, red car obj_4, tree obj_6. ✗

DepthEvidence: traffic light obj_1, person in purple jacket obj_5, shopping cart_2 obj_3, shopping cart_1 obj_2, building pillar obj_7, red car obj_4, tree obj_6. ✓

Metric Depth

Metric Pair.



metric_gap_threshold; box prompt

Question. Is the average camera depth of wooden entertainment center obj_2 at least 6.60 m greater than that of blue sofa obj_1?

Ground truth: No.

Gemini-3.1-Pro: yes. ✗

DepthEvidence w/o Interface: no. ✓

DepthEvidence: No. ✓

Spatial Metric Pair.



spatial_metric_gap; box prompt

Question. What is the absolute difference in average camera depth between the object to the right of black sedan obj_1 and building obj_3?

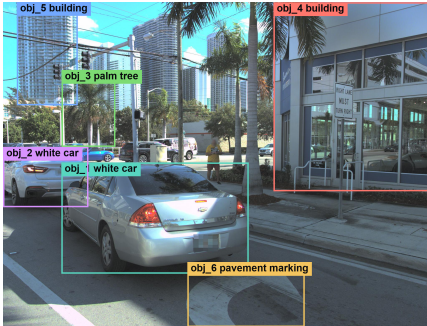
Ground truth: 38.39 m

Gemini-3.1-Pro: 10.5 ✗

DepthEvidence w/o Interface: 27.00 ✗

DepthEvidence: 40.00 m ✓

Spatial Metric Group.



spatial_range_filter; box prompt

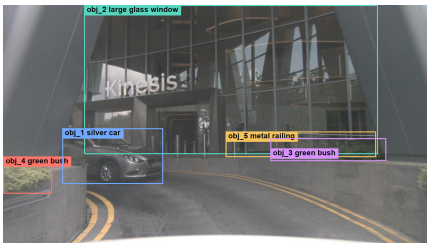
Question. Among the referenced objects to the left of building obj_4, which have an average camera depth between 55.21 m and 79.57 m, inclusive?

Ground truth: building obj_5.

Gemini-3.1-Pro: obj_3 ✗

DepthEvidence w/o Interface: building obj_5. ✓

DepthEvidence: building obj_5. ✓



compositional_metric_constraint; box prompt

Question. Among the referenced objects to the right of green bush obj_4, select those with an average camera depth greater than 4.50 m but smaller than that of large glass window obj_2.

Ground truth: silver car obj_1, metal railing obj_5, green bush obj_3.

Gemini-3.1-Pro: obj_3, obj_5 ✗

DepthEvidence w/o Interface: silver car obj_1, green bush obj_3, metal railing obj_5. ✓

DepthEvidence: silver car obj_1, metal railing obj_5, green bush obj_3. ✓

Metric Set Decision.



metric_margin_decision; box prompt

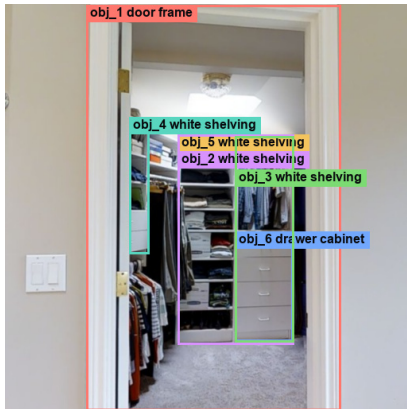
Question. Select the closest referenced object only if it is at least 21.00 m closer in average camera depth than every other candidate; otherwise answer ambiguous.

Ground truth: ambiguous.

Gemini-3.1-Pro: obj_1 ✗

DepthEvidence w/o Interface: asphalt road obj_1. ✗

DepthEvidence: ambiguous. ✓



camera_safety_band; box prompt

Question. Is any referenced object at an average camera depth of no more than 0.80 m?

Ground truth: No.

Gemini-3.1-Pro: yes ✗

DepthEvidence w/o Interface: no ✓

DepthEvidence: No. ✓